

BINYAN XU

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Education

The Chinese University of Hong Kong

Ph.D. Student in Information Engineering; Advisor: Prof. Kehuan Zhang

Sep. 2023 – Oct. 2027 (expected)

GPA: 3.914

University of California, Berkeley

Exchange Program in EECS

Aug. 2021 – Dec. 2021

GPA: 4.0

Xi'an Jiaotong University

B.Sc. in Automation, Qian Xuesen College; Qian Xuesen Honors Program

Sep. 2019 – Jul. 2023

GPA: 3.92; Rank: Top 5%

- Entered XJTU from junior high via the Honors Youth Program (100 admitted from 4,000+ applicants).

Research Thesis

- One thesis: closed-loop intelligence from risk sensing to reliable autonomy**—letting intelligent systems sense states and risks in open environments, accumulate memory and skills, and achieve reliable, low-cost long-horizon autonomy via external verification and risk feedback.
- The same “**discover failures** → **verify externally** → **optimize autonomy**” method drives my work: early on I discovered and verified AI failure modes in multimodal and system settings; recently I moved to state management, memory, and skill distillation for long-horizon LLM agents.
- First/co-first author on 4 accepted and 4 under-review CCF-A works, spanning long-horizon LLM agents, multimodal foundation models, and AI risk; strong in system building, benchmarks/ablations, and artifact release.

Selected Publications and Manuscripts

4 accepted and 4 under review at CCF-A venues; all are first/co-first-author works.

I. Long-Horizon Autonomous Agents: State, Memory, and Skills

[1] **LLM Agents Are Latent Context Managers: Eliciting Self-Managed Context via Dashboard**  Under Review

Binyan Xu, Haitao Li, Kehuan Zhang.

State awareness: enables LLM agents to sense and manage their own working state for long-horizon tasks.

[2] **From Multi-Agent to Single-Agent: When Is Skill Distillation Beneficial?**  Under Review

Binyan Xu, Dong Fang, Haitao Li, Kehuan Zhang.

Skill consolidation: identifies when multi-agent collaboration/evaluation loops can be compressed into cheaper single-agent skills.

[3] **Contextual Agentic Memory is a Memo, Not True Memory**  Under Review

Binyan Xu*, Xilin Dai*, Kehuan Zhang. *Equal contribution.

Memory maintenance: separates temporary context retrieval from reusable experience, redefining agent memory.

II. External Auditing and Semantic Verification: Feedback Loops for Untrusted Models

[4] **From Internal Diagnosis to External Auditing: A VLM-Driven Paradigm for Online Defense**  ICML 2026

Binyan Xu, Xilin Dai, Fan Yang, Di Tang, Kehuan Zhang.

Poster

External auditing: uses an external VLM to audit backdoored predictions without trusting the deployed model itself.

[5] **CLIP-Guided Backdoor Defense through Entropy-Based Poisoned Dataset Separation**  ACM MM 2025

Binyan Xu, Fan Yang, Xilin Dai, Di Tang, Kehuan Zhang.

 Oral Presentation

Semantic verification: uses CLIP signals to separate poisoned images and remove hidden backdoor behavior.

III. Risk Discovery and Feedback: From Model Vulnerabilities to Automation Decisions

[6] **One Surrogate to Fool Them All: Universal and Transferable Adversarial Attacks with CLIP**  CCS 2025

Binyan Xu, Xilin Dai, Di Tang, Kehuan Zhang.

 Oral Presentation

State transfer: leverages CLIP’s multimodal semantic representation to craft transferable attacks.

[7] **Breaking Stealth-Potency Trade-off in Clean-Image Backdoors with Trigger Optimization**  AAAI 2026

Binyan Xu, Xilin Dai, Fan Yang, Di Tang, Kehuan Zhang.

 Oral Presentation

Risk discovery: reveals how generative AI can create subtle training-set contamination that bypasses existing defenses.

[8] **When Agent Automation Becomes Profitable: Quantify and Insure Autonomous AI Risk**  Under Review

Binyan Xu, Xilin Dai, Fan Yang, Kehuan Zhang.

Risk feedback: models when agent automation is economically worthwhile under failure cost and insurance-style pricing.

Experience

Closed-Loop Autonomous Agents: State, Memory, and Skills

Jan. 2026 – May 2026

Tencent IEG, Lightspeed Studios, X-Data Research Team; Qingyun Program Internship

Full-time Research Intern

- Reframed long-horizon agents as a systems problem of **state awareness, memory management, skill internalization, and cost control**, rather than merely enlarging context windows or stacking multi-agent workflows.
- Led *LLM Agents Are Latent Context Managers* [1], identifying missing *proprioception* over an agent’s own context state and designing the VISTA self-managed context layer with typed/addressable memory blocks, token usage, recency, access history, and budget exposure.
- Improved Gemini-3-Flash success rate on 75 long-horizon tool-use tasks in LOCA-Bench from 22.7% to 50.7%, outperforming ReAct, SLIM, Active Context Compression, and Claude Code; observed gains across 4 backbones.
- Led skill-distillation research [2] introducing *Metric Freedom* to predict when multi-agent collaboration/evaluation structures can be internalized as single-agent skills; matched or exceeded original systems at 1.4–15× lower cost across 4 tasks and 11 datasets.
- Co-led agentic-memory research [3] showing that many current “memory” modules are context extensions rather than true memory systems; built evaluation, ablation, statistical analysis, and reproducible artifacts toward reusable agent experience.

Controllable Generative AI for Graphic Design

Jul. 2022 – Jun. 2023

Microsoft Research Asia, Data, Knowledge and Intelligence; Stars of Tomorrow Internship Program

Full-time Research Intern

- Co-developed an AI-aided aesthetic design system combining layout planning, vision-language understanding, and diffusion-based style/decorative-pattern generation for controllable content creation in real design workflows.
- Implemented controllable generation modules that improved visual quality while reducing layout, element-relation, and design-constraint violations, providing application-side experience in controllable generation, constraint satisfaction, and human-feedback loops.
- Received the *Award of Excellence* in the MSRA Stars of Tomorrow Internship Program for research contribution and full-time participation.

Closed-Loop Trustworthy AI: Auditing, Risk Discovery, and Verification

Sep. 2023 – Jan. 2026

Laboratory for Applied Security Research, CUHK

Ph.D. Research

- Positioned security research as **risk sensing and external verification** in closed-loop trustworthy AI systems, covering VLM-based external auditing, multimodal-LLM-guided poisoned data separation, CLIP-based adversarial transfer, generative clean-image backdoors, and autonomous AI risk analysis [4–8].
- Proposed data-free online backdoor defense using VLMs as external auditors [4], shifting defense from internal neural diagnosis to externally grounded semantic inspection for settings where deployed models cannot be fully trusted.
- Designed universal trigger optimization methods [7] that achieve high success rates against real AI services, including Google Image Search, Baidu Image Search, Taobao Image Search, and multimodal models such as GPT-4 and Claude Sonnet, characterizing transfer risks from open foundation models.
- Built CLIP-guided poisoned-dataset-separation and attack pipelines [5,6]; produced accepted/submitted CCF-A papers at ICML, AAAI, CCS, and ACM MM, forming a continuous trajectory from risk discovery to external semantic verification.

Open-Source & Impact

- Open-sourced first-author research code at github.com/binyxu, covering long-horizon agent evaluation, VISTA self-managed context, agentic memory, skill distillation, VLM-based external auditing, backdoor attack/defense, and adversarial transfer.
- Released reproducible implementations and artifacts that turn papers into reusable benchmarks, agent harnesses, auditing workflows, and attack/defense pipelines for community verification and follow-up development.
- Agent-memory work received broad community discussion after release, including coverage by DAIR.AI on X (26K+ views and 700+ likes/bookmarks), Facebook AI communities, YouTube explainers, Xiaohongshu posts, and Xinzhiyuan coverage.

Honors & Skills

- **Academic service:** Official reviewer for ICML, ICLR, ACM MM, and AAAI; oral presentations at AAAI, ACM CCS, and ACM MM; poster presentation at ICML; Lead TA mentoring 40+ undergraduate final-year project students.
- **Honors:** Finalist Award, Mathematical Contest in Modeling (Top 2%); selected into XJTU Honors Youth Program (Top 2.5% among 4,000+ applicants); Qian Xuesen Honors Program Graduate / Outstanding Graduate; first-class funding for Berkeley Exchange Program; MSRA Stars of Tomorrow Award of Excellence.
- **Research:** closed-loop autonomous intelligence, long-horizon LLM agents, agentic memory, self-managed context, self/skill distillation, agent automation/risk, multimodal foundation models, AI safety/security.
- **Programming and Tools:** Python (primary); experienced with PyTorch, Hugging Face, vLLM, OpenAI API, and agent harnesses, tool-use evaluation, ablation pipelines, and multi-GPU experimentation (up to 16 GPUs).
- **Languages:** Chinese (native), English (fluent; TOEFL 104).